

Multiplicity Theory v0.2: A Self-Correcting Functorial Framework

Abstract

We formalize **Multiplicity** as a functorial, semiring-valued invariant on a symmetric monoidal $(\infty, 1)$ -category endowed with descent and a domain-specific **self-correction** operator. The framework recovers classical algebraic–geometric multiplicities, extends to spectral and network settings, and provides stability windows under perturbations.

1. Core Formalization

1.1 Minimal Object & Target

Let: - $(\mathcal{C}, \oplus, \otimes)$ be a symmetric monoidal $(\infty, 1)$ -category with a Grothendieck topology J . - $\mathbf{Rig}$ be the 2-category of commutative semirings and semiring functors. - $\mathbf{Mult} : (\mathcal{C}, \oplus, \otimes) \rightarrow (\mathbf{Rig}, +, \cdot)$ be a **lax symmetric monoidal functor**, with base-change natural in objects. Write $\mathbf{m}(X) \in \mathbb{M}_X$ for its value.

1.2 Axioms (Essential)

A1 (Functoriality): $\mathbf{Mult}(g \circ f) = \mathbf{Mult}(g) \circ \mathbf{Mult}(f)$.

A2 (Semiring Additivity/Multiplicativity):

$$\mathbf{m}(X \oplus Y) = \mathbf{m}(X) + \mathbf{m}(Y), \quad \mathbf{m}(X \otimes Y) = \mathbf{m}(X) \cdot \mathbf{m}(Y).$$

A3 (Descent): $\mathbf{Mult}$ is a J -sheaf of semirings (Čech descent holds for admissible covers).

A4 (Invariance): Preserved under relevant equivalences (isomorphism, derived, Morita, homotopy).

A5 (Normalization): Restricts to classical multiplicities on canonical subcategories (Hilbert–Samuel/ ℓ , Serre/derived intersection, tropical stable intersections, spectral degeneracy, microstate count in statistical mechanics).

A6 (Derived Additivity): For nontransverse intersections/gluing, A2 holds after **derived correction** (Tor/Ext terms or obstruction-theoretic defects).

A7 (Self-Correction): There exists a domain-specific update operator $\mathcal{U}$ and a metric d with **contractivity**:

$$d(\mathbf{m}_{t+1}, \mathbf{m}^*) \leq \kappa d(\mathbf{m}_t, \mathbf{m}^*), \quad 0 \leq \kappa < 1.$$

1.3 Exclusion Principle

An admissible **Mult** must be: 1. **Integer-valued** on compact discrete presentations. 2. **Localizable** (computable from a cover with descent). 3. **Stable** under the domain's equivalence notion.

This excludes Shannon/von Neumann entropy as a universal substitute (they generally fail (1) and/or (3) across categories).

2. Self-Correction: Operators & Stability Windows

We detail **objects**, **multiplicity**, **update law**, and **stability** for three anchor domains.

2.1 Quantum/Spectral (Q)

- **Object:** Self-adjoint H with spectral projectors $\{P_\lambda\}$.
- **Multiplicity:** $\mathbf{m}(H) = \{(\lambda, \text{rank } P_\lambda)\}$ or cluster-multiplicity after gap-based clustering.
- **Update:** $H \mapsto H + E$ (bounded Hermitian perturbation) or CPTP channel $\mathcal{E}$.
- **Stability (Davis-Kahan):** If $\|E\| < \frac{1}{2} \text{gap}$, then projector angles satisfy

$$\sin \Theta(P, P') \leq \frac{\|E\|}{\text{gap}} =: \kappa < 1,$$

and the cluster multiplicity is locally constant.

Metric choice: principal-angle metric on invariant subspaces; cluster stability judged by rank constancy within the gap window.

2.2 Algebraic/AG (A)

- **Object:** Noetherian local pair $(R, \mathfrak{m}; I)$ (or maps $X \rightarrow Y$).
- **Multiplicity:** Hilbert-Samuel $e_I(M)$; derived intersection multiplicity via $\sum (-1)^i \text{length Tor}_i$.
- **Update:** **Deformation to the normal cone**; flat base change.
- **Stability:** Upper semicontinuity of e in flat families $\Rightarrow e_{t+1} \leq e_t$ for flattening updates (one-sided contraction in the partial order).

Metric choice: partial order $\geq$ on multiplicities; optionally $|e_{t+1} - e^*|$ in regimes where equality is expected.

2.3 Graphs/Networks (G)

- **Object:** Finite graph G on n vertices.
- **Multiplicity:** Coefficient vector $\mathbf{a}(G) = (a_0, \dots, a_m)$ of the **reliability polynomial** $R_G(p) = \sum_k a_k p^k (1-p)^{m-k}$.
- **Update:** **Deletion-contraction** recursion: $R_G = pR_{G/e} + (1-p)R_{G \setminus e}$; edit operators add/delete edges.

- **Stability:** Coefficients obey submodularity; under $\leq r$ edge edits,

$$\|\mathbf{a}(G) - \mathbf{a}(G')\|_1 \leq c r n^{O(1)}.$$

Metric choice: ℓ^1 distance on coefficient vectors; log-summaries for robustness proxies.

3. Universal Constructions

Multiplicity Sheaf. A presheaf $\mathcal{M} : \mathcal{O}(X)^{op} \rightarrow \mathbf{Rig}$ that satisfies descent, gluing local multiplicities into $\mathbf{m}(X)$.

Multiplicity Monad. A monad $T : \mathcal{C} \rightarrow \mathcal{C}$ encoding branching/superposition; the Kleisli category composes multiplicity-bearing processes, compatible with A2 and A3.

4. Explicit Examples & Counterexamples

1. **Monomial Ideals** (toy algebraic backend): in $k[x, y]$ at $\mathfrak{m} = (x, y)$, $I = (x^a, y^b)$ has $\mathbf{m} = e_{\mathfrak{m}}(R/I) = ab$. Verifies A2 across random (a, b) .
 2. **Derived Additivity Toy:** $C_1 : y - x^2$, $C_2 : y$ intersect nontransversely at 0. Naive additivity fails; Tor-corrected multiplicity satisfies A6.
 3. **Spectral Robustness:** $H_0 = \text{diag}(0, 0, \Delta)$ with perturbation εE . The 0-eigenspace multiplicity is stable for $\varepsilon < \Delta/2$ and changes at gap closure.
 4. **Graph Non-Entropy Witness:** Two 8-vertex graphs with the same degree sequence yield distinct reliability coefficients; $\mathbf{m}$ distinguishes them while degree-entropy does not.
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5. Micro-Theorems (Targets)

Thm 1 (Descent). If $X = U \cup V$ is a J -cover with derived-transverse $U \cap V$, then

$$\mathbf{m}(X) = \mathbf{m}(U) + \mathbf{m}(V) - \mathbf{m}(U \cap V).$$

Thm 2 (Monoidal). For independent systems X, Y , $\mathbf{m}(X \otimes Y) = \mathbf{m}(X) \cdot \mathbf{m}(Y)$.

Thm 3 (Spectral Stability Window). Let H have a spectral cluster separated by gap g . If $\|E\| < g/2$, cluster multiplicity is invariant and projector distance contracts by factor $\|E\|/g$.

Prop 4 (Graph Lipschitz). If G' differs by $\leq r$ edge edits, then $\|\mathbf{a}(G) - \mathbf{a}(G')\|_1 \leq c r n^{O(1)}$.

Prop 5 (Upper Semicontinuity). In flat families, Hilbert–Samuel multiplicity is upper semicontinuous; flattening updates are contractive in the partial order $\geq$.

6. API Skeleton (Language-Agnostic)

```
class Mult[Value: Semiring]:
  value: Value
  def op_sum(a: "Mult", b: "Mult") -> "Mult": ...
  def op_prod(a: "Mult", b: "Mult") -> "Mult": ...
  def transport(self, f: Morphism) -> "Mult": ... # functorial transport

interface Backend[X, Value: Semiring]:
  def compute(x: X) -> Mult[Value]
  def local(self, U: CoverOpen[X]) -> Mult[Value] # optional, for sheaf
  gluing

namespace backends:
  algebraic: Backend[(R, m; I), N]
  tropical: Backend[TropicalCycle, N]
  spectral: Backend[Matrix | Hamiltonian, N]
  graphs: Backend[Graph, PolyRing]

module update:
  def spectral_step(H: Matrix, E: Matrix) -> Mult[N]: ...
  def deform_step(instance, deformation) -> Mult[N]: ...
  def graph_step(G: Graph, edit) -> Mult[PolyRing]: ...
```

7. Experiments & Datasets (Week-1 Plan)

Algebraic: Grid $(a, b) \in \{2..8\}^2$ for $I = (x^a, y^b)$; verify A2/A3; random deformations $y^b \mapsto y^b + x^c$.

Spectral: Random Hermitian 3×3 and 6×6 with planted gaps; sweep $\|E\|$ to map stability windows.

Graphs: $ER(n, p)$ and $BA(n, m)$ with $n = 50..300$; correlate percolation threshold with (i) degree entropy, (ii) algebraic connectivity, (iii) log of reliability coefficients.

Success criteria: Axioms hold on toy cases; one domain where **m** predicts behavior that entropy/rank miss; reproducible notebooks demonstrate stability.

8. Deliverables & Links

Paper skeleton (LaTeX):

- [multiplicity_v0.2.tex](#)

Code scaffold (`multiplicity_core`):

- [README.md](#)
- [pyproject.toml](#)
- [src/multiplicity_core](#)
- [tests](#)

Demos (Jupyter):

- [spectral_stability.ipynb](#)
 - [graph_reliability_vs_robustness.ipynb](#)
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9. Roadmap (Next 2–3 Weeks)

1. Prove Thm 1–3 on restricted settings; add counterexamples clarifying A6 necessity.
 2. Implement transport along morphisms (functoriality) for each backend.
 3. Add tropical backend (stable intersections).
 4. Integrate spectral gap detection and explicit Davis–Kahan bounds visualization.
 5. Prepare preprint with toy benchmarks and public repo.
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10. Open Problems & Extensions

- Characterize the broadest class of semirings $\mathbb{M}$ admitting A1–A7 simultaneously.
- Relate **Mult** to K -theory / λ -rings; identify when **m** refines ranks/Chern characters.
- Extend to probabilistic sheaves and Wasserstein contractivity for stochastic updates.
- Noncommutative geometry: Morita-invariant multiplicities beyond classical spectral degeneracy.
- Data-driven multiplicities: learnable base-change functors respecting A1–A4.